

GeoPKI

I. MODEL

The earth ellipsoid model of the existing world geodetic system (WGS) 84 [1] is used. WGS 84 defines the *semi-major axis* (half of the largest diameter of an ellipsoid, at the equator) a to be 6'378'137.0 meters [1]. The *semi-minor axis* (the polar radius) b is defined as 6'356'752.3142 meters [1]. Standard polar coordinates (longitude, latitude) are used for referencing points on the planet's two-dimensional surface and a third coordinate, elevation, extends this to three dimensions. Longitude, latitude and elevation together refer to a unique point in three-dimensions.

Each of the three coordinates is discretized in a way that preserves meter-precision in the worst case. With the semi-major axis a , the maximum circumference of the ellipsoid equals $C_x := 2a\pi$ meters. Using base two, $B_x := \lceil \log_2(C_x) \rceil = 26$ bits for the longitude thus suffice to represent each possible meter value in the worst case. For the latitude we have $C_y := 2b\pi$ meters and thus $B_y := \lceil \log_2(C_y) \rceil = B_x =: B_{x,y}$, the same minimum number of bits are required. For elevation there is no natural bound so one has to be fixed. At the time of writing, the highest point on earth is below 10'000 meters [] and the lowest layer of earth's atmosphere, the troposphere, extends to about 18 km [2]. Thus a limit of $H = 20'000$ m and thus $B_z := \lceil \log_2(H) \rceil = 15$ bits seems reasonable.

II. DATA STRUCTURE

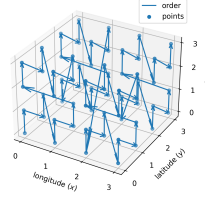
Inspired by F-PKI [3], sparse merkle hash tree (MHT) is used. Each leaf represents a full discretized coordinate using $B := 2 \cdot B_{x,y} + B_z = 68$ bits resulting in a total of 2^B leaves. To have spatially close points somewhat close together in the MHT, the space-filling Z-order curve is used to sort the three dimensional locations in one dimension. An example with $B_{x,y} = B_z = 2$ is shown in Figure 1a. GeoPKI uses a slightly modified 3D Z-order curve: Rather than using the full 3D Z-order curve, a 2D Z-order curve over the longitude and latitude is used and points with the same longitude and latitude but different elevation are put in sequence. While this leads to spatially close points being further apart in the 1-dimensional sorting, this modification allows for shorter proofs in a majority of envisioned use-cases (will be discussed in Section II-B) and even results in a nicer binary representation. Figure 1b contrasts this modification with the standard 3D Z-order curve shown in Figure 1a.

A. Leaf Binary Format

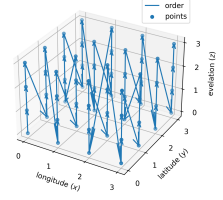
Each leaf of the MHT corresponds to a point and the leaves are ordered by the binary representation of that point. A point tuple $p := (x, y, z)$ at longitude x , latitude y and elevation z with

$$(x, y, z) \in [-180, 180] \times [-90, 90] \times [0, H]$$

is first discretized to



(a) 3D Z-Order Curve



(b) 2D Z-Order Curve with elevation in sequence

Fig. 1: Difference between a standard 3D Z-order curve and the one used by GeoPKI when using two bits for all coordinates.

$$p' := \left(\left\lfloor \frac{x + 180}{360} \cdot C_x \right\rfloor, \left\lfloor \frac{y + 90}{180} \cdot C_y \right\rfloor, \lfloor z \rfloor \right)$$

so that the following folds:

$$p' \in [0, 2^{B_x}) \times [0, 2^{B_y}) \times [0, 2^{B_z})$$

This formula holds since the first term in the longitude and latitude tuples can be at most 1 and since $C_x < 2^{B_{x,y}}$, $C_y < 2^{B_{x,y}}$ and $H < 2^{B_z}$ hold. The three resulting integer numbers can now be represented using $B_{x,y}$, $B_{x,y}$ and B_z bits, respectively. Hence let $p'' := (x_b, y_b, z_b)$ with

$$p'' \in \{0, 1\}^{B_{x,y}} \times \{0, 1\}^{B_{x,y}} \times \{0, 1\}^{B_z}$$

For $c \in \{x, y, z\}$ and for any integer $i \in [0, 2^{B_c})$, let $c_{b,i}$ denote the i -th bit of c_b . Moreover let $\parallel$ denote the concatenation operator. The binary representation p_b of the point p is then defined as

$$p_b = \left(\bigg\|_{i=0}^{B_{xy}-1} x_{b,i} \parallel y_{b,i} \right) \parallel \left(\bigg\|_{i=0}^{B_z-1} z_{b,i} \right)$$

In words, the binary representation of p is the interleaving of the binary representation for the longitude and latitude concatenated to the binary representation of the elevation.

B. Querying the MHT

Each intermediate MHT node is a prefix of its children To further reduce the required storage space, intermediate nodes in the MHT are allowed to contain certificates. This allows

C. Proof of Inclusion

To proof a given

D. Proof of Absence

III. OPTIMIZATIONS

REFERENCES

- [1] NGA geomatics - WGS 84. [Online]. Available: <https://earth-info.nga.mil/?dir=wgs84&action=wgs84>

- [2] S. K. Dhaka and V. Kumar, "Chapter 1 - composition and thermal structure of the earth's atmosphere," in *Atmospheric Remote Sensing*, ser. Earth Observation, A. Kumar Singh and S. Tiwari, Eds. Elsevier, pp. 1–18. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/B9780323992626000237>
- [3] L. Chuat, C. Krähenbühl, P. Mittal, and A. Perrig, "F-PKI: Enabling innovation and trust flexibility in the HTTPS public-key infrastructure," in *Proceedings 2022 Network and Distributed System Security Symposium*. [Online]. Available: <http://arxiv.org/abs/2108.08581>